IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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- Introduction
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Executive Summary

SpaceX has revolutionized space travel by significantly reducing launch costs through reusability. With launch prices at $62M—compared to over $165M for competitors—SpaceX achieves cost efficiency primarily by landing and reusing the Falcon 9's first stage. Accurately predicting the success of this landing is essential for stakeholders assessing mission costs or competing for launch contracts.

This project applies data science and machine learning techniques to predict the probability of first-stage landings. Using historical launch data, a comprehensive ML pipeline was developed to extract insights and support predictive modeling.

Methodology Overview:

- **Data Acquisition & Cleaning:** Gathered and preprocessed SpaceX launch data from public APIs and previous labs.
- **Feature Engineering and Standardization:** Created key features such as payload mass, orbit type, launch site, and booster version.
- **Exploratory Data Analysis (EDA):** Identified correlations and trends in successful landings.
- **Model Selection & Training:** Applied classification models including Logistic Regression, SVM, Decision Trees, and K-Nearest Neighbors (KNN).
- **Hyperparameter Tuning:** Optimized models using GridSearchCV and cross-validation.
- **Model Evaluation:** Assessed performance using accuracy, precision, recall, and F1 score.

Prediction Analysis Results (Classification):

- Classification models were compared and evaluated.
- The decision tree was the best-performing model, achieving a test accuracy of 88.89% in predicting successful landings.

Introduction

Project Background & Problem Statement

SpaceX offers Falcon 9 launches for $62M, much cheaper than competitors charging $165M+. This cost advantage largely comes from reusing the first stage of the rocket.

Knowing whether the first stage lands can help estimate launch costs—valuable for competitors looking to bid against SpaceX.

Goal

Build a machine learning model to predict first-stage landing success using launch data by:

- Performing EDA
- Creating a classification pipeline
- Identify the best performing algorithm among:
    - Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbor(KNN) Algorithm.
- Selecting the best model based on test accuracy

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Data Acquisition: Launch data collected from public SpaceX APIs and CSV files
  - Data Wrangling: Cleaned and structured data using pandas; handled missing values using mean imputation
  - Data Processing: Engineered features such as payload mass, launch site, orbit type, and booster version to support modeling
- Perform exploratory data analysis (EDA) using visualization and SQL
  - EDA with Visuals & SQL: Used Seaborn, Matplotlib, and SQL queries to uncover key insights and correlations
  - Interactive Dashboards: Built geographic and temporal visualizations using Folium and Plotly Dash for dynamic exploration of launch outcomes

Methodology

Executive Summary

- Perform predictive analysis using classification models
  - Classification Models: Applied Logistic Regression, SVM, Decision Trees, and K-Nearest Neighbors to predict landing success
  - Model Tuning: Optimized models using GridSearchCV and cross-validation techniques
  - Evaluation Metrics: Assessed performance using accuracy, precision, recall, and F1-score for model comparison and selection

Data Collection

- SpaceX API
  - Used to collect structured data on launches, rockets, payloads, and landing outcomes.
- Wikipedia (Falcon 9 Launch List)
  - Scraped using BeautifulSoup to get additional launch history and mission details.
- NASA/Open Data
  - Provided launch site coordinates and location info.
- Python Tools Used
  - requests, BeautifulSoup, pandas
- All data was cleaned and merged into a single DataFrame for analysis.

Data Collection – SpaceX API

- API Request – Access SpaceX launch data using a static JSON endpoint.Response
- Validation – Confirm data retrieval with a 200 OK status code.
- Data Parsing – Convert the JSON response to a flat table using pd.json_normalize.

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

In [9]:
```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_
```

We should see that the request was successfull with the 200 status response code

In [10]:
```
response=requests.get(static_json_url)
```

In [11]:
```
response.status_code
```

Out[11]: 200

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

In [12]:
```
# Use json_normalize meethod to convert the json result into a dataframe
data = pd.json_normalize(response.json())
```

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%201%20Part%201-%20Data%20Collection%20API.ipynb

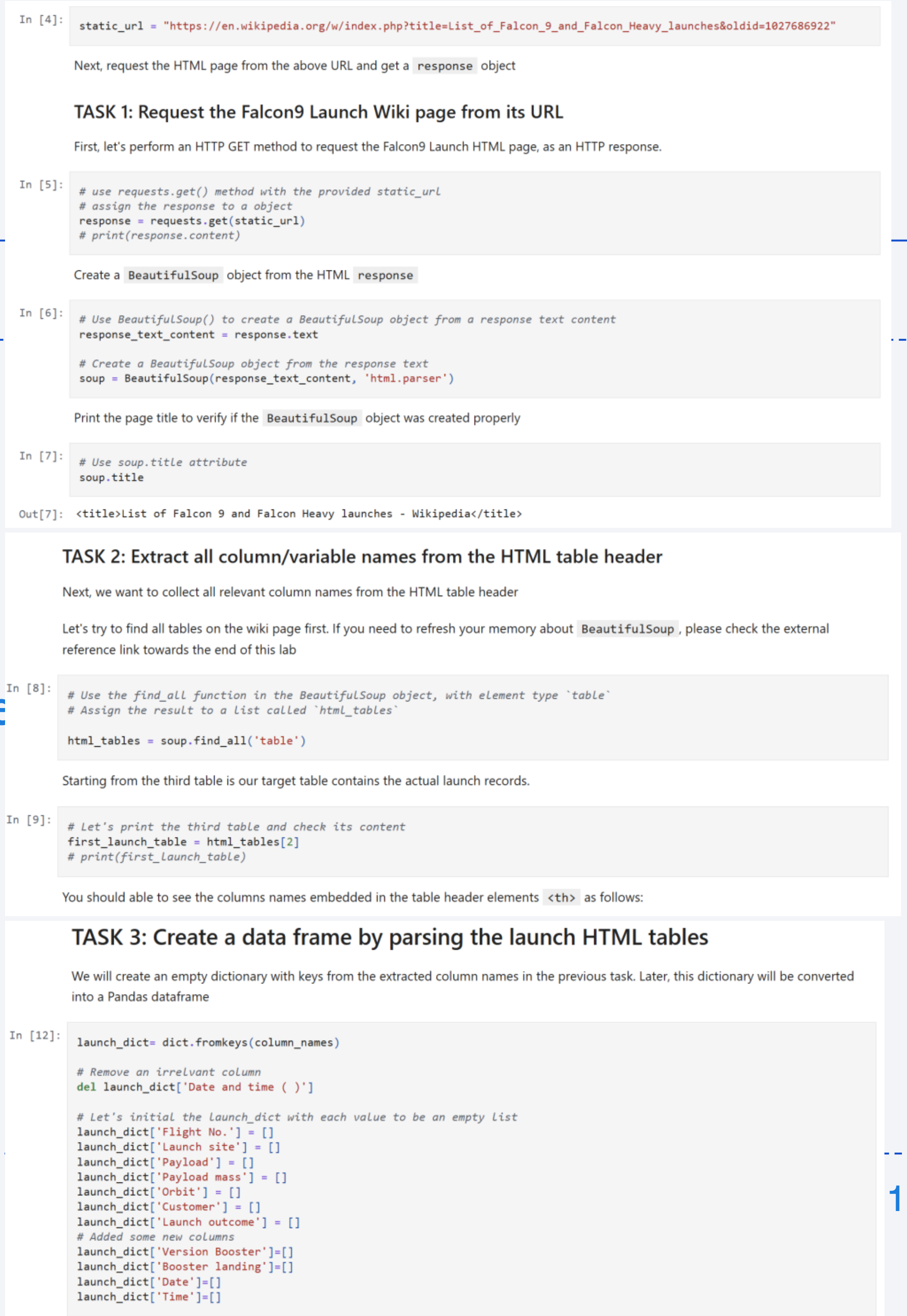
Data Collection - Scraping

Web Scraping Process Using **requests** & **BeautifulSoup**

1. Import requests and **BeautifulSoup**
2. Fetch page with **requests.get()**
3. Parse HTML using **BeautifulSoup**
4. Find all tables: **soup.find_all('table’)**
5. Select target table (3rd table)
6. Extract headers from **<th>**
7. Create and clean column dictionary

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%201%20Part%202-%20Web%20Scraping.ipynb

Place



In [4]:
```
static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"
```

Next, request the HTML page from the above URL and get a response object

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

In [5]:
```
# use requests.get() method with the provided static_url
# assign the response to a object
response = requests.get(static_url)
# print(response.content)
```

Create a BeautifulSoup object from the HTML response

In [6]:
```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content
response_text_content = response.text

# Create a BeautifulSoup object from the response text
soup = BeautifulSoup(response_text_content, 'html.parser')
```

Print the page title to verify if the BeautifulSoup object was created properly

In [7]:
```
# Use soup.title attribute
soup.title
```

Out[7]:
```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about BeautifulSoup, please check the external reference link towards the end of this lab

In [8]:
```
# Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`

html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

In [9]:
```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
# print(first_launch_table)
```

You should able to see the columns names embedded in the table header elements <th> as follows:

TASK 3: Create a data frame by parsing the launch HTML tables

We will create an empty dictionary with keys from the extracted column names in the previous task. Later, this dictionary will be converted into a Pandas dataframe

In [12]:
```
launch_dict= dict.fromkeys(column_names)

# Remove an irrelvant column
del launch_dict['Date and time ( )']

# Let's initial the launch_dict with each value to be an empty list
launch_dict['Flight No.'] = []
launch_dict['Launch site'] = []
launch_dict['Payload'] = []
launch_dict['Payload mass'] = []
launch_dict['Orbit'] = []
launch_dict['Customer'] = []
launch_dict['Launch outcome'] = []
# Added some new columns
launch_dict['Version Booster']=[]
launch_dict['Booster landing']=[]
launch_dict['Date']=[]
launch_dict['Time']=[]
```

Data Wrangling

Data wrangling is essential to transform raw, unstructured data into a clean and structured format. This process enables accurate analysis, reveals key patterns, and ensures reliable labels for training supervised machine learning models.

- Launches per Site
  - Counted number of launches at each launch site
- Orbit Frequency
  - Counted occurrences of each orbit type
- Mission Outcomes by Orbit
  - Counted success/failure for each orbit
- Landing Outcome Label
  - Created binary label (1 = success, 0 = failure) from mission outcome descriptions

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%201%20Part%203%20-%20Data%20Wrangling.ipynb

EDA with Data Visualization

- Categorical Scatter Plots
  - Visualized FlightNumber, PayloadMass, and LaunchSite against landing Class (success/failure) to identify direct relationships.
- Bar Plot
  - Compared landing success rates across Orbit types.
- Line/Scatter Plots
  - Showed Class trends over FlightNumber, PayloadMass, and Date for BoosterVersion and LaunchSite to observe performance evolution.

Overall Purpose: These visualizations were used for Exploratory Data Analysis (EDA) to identify key factors and trends influencing the Falcon 9 first stage landing success.

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%202%20Part%202-%20EDA%20with%20Visualization%20Lab.ipynb

EDA with SQL

- Retrieved all unique launch sites.
- Filtered first 5 launches from sites beginning with 'CCA'.
- Calculated total payload mass by NASA (CRS).
- Found average payload mass for booster version 'F9 v1.1'.
- Identified the first successful landing on a ground pad.
- Listed booster versions with successful drone ship landings and payload between 4000–6000 kg.
- Counted total mission outcomes grouped into success and failure.
- Used a subquery to find booster versions with the maximum payload mass.
- Extracted data for failed drone ship landings in 2015 showing month names and relevant details.
- Ranked landing outcomes by frequency for missions between June 2010 and March 2017.

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%202%20Part%201-%20EDA%20SQL%20SQLlite.ipynb

Build an Interactive Map with Folium

- Mark all launch sites on the map and visually display their locations with circles.
  - Clearly mark and visualize each launch site's location on the map.
- Use green or red markers to indicate success or failure.
  - Show launch outcomes at a glance—green for success, red for failure.
- Add the mouse position to obtain launch site coordinates.
  - Lets users see coordinates anywhere on the map for exploration or analysis.
- Add a polyline to measure the distance from a launch site to the coastline.
  - Visualize the distance from each launch site to the nearest coastline.
- Calculate the distance between a launch site and its neighbors
  - Helps analyze spatial relationships and logistics between sites.

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%203%20Part%201%20-%20Interactive%20Visual%20Analytics%20with%20Folium%20lab.ipynb

Build a Dashboard with Plotly Dash

- Launch Site Dropdown: Select a specific launch site or view "All Sites.“
  - Reason: Enables focused analysis by location.
- Success Pie Chart: Displays launch success rates. Shows success by site (All Sites) or success vs. failure (specific site).
  - Reason: Visualizes launch performance.
- Payload Range Slider: Filters data by payload mass range.
  - Reason: Allows investigation of payload mass impact.
- Payload vs. Success Scatter Chart: Shows the relationship between payload, booster version, and launch success, filtered by site and payload range.
  - Reason: Identifies correlations between payload, booster, and outcome.Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%203%20Part%202%20-%20Build%20an%20Interactive%20Dashboard%20with%20Plotly%20Dash.py

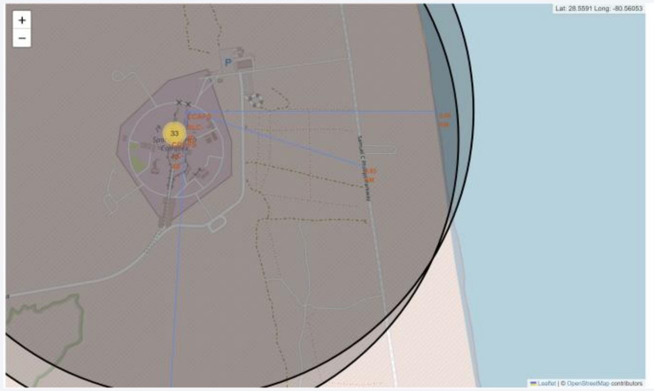
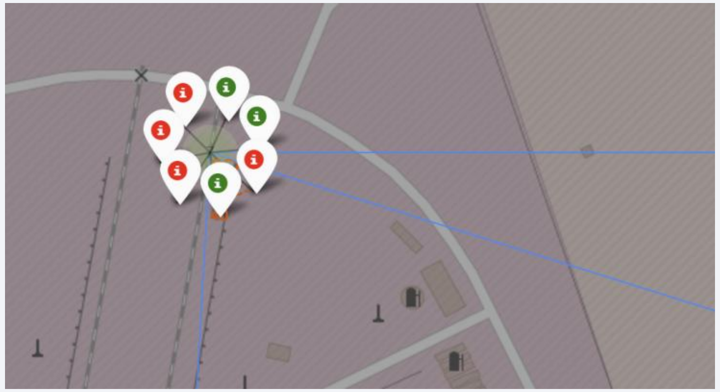
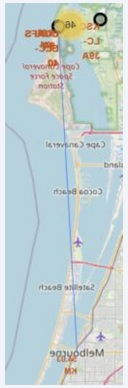
Predictive Analysis (Classification)

1. Build & Prepare:
   - Loaded preprocessed data (X, Y).
   - Standardized features (X).
   - Split data into training (80%) and test (20%) sets.
2. Improve & Evaluate (Training):
   - Trained four models: Logistic Regression, SVM, Decision Tree, KNN.
   - Improved each via GridSearchCV (10-fold CV) to find optimal hyperparameters.
   - Evaluated validation accuracy during tuning.
3. Evaluate (Testing) & Find Best:
   - Assessed final model performance on unseen test data.
   - Used test accuracy and confusion matrices.
   - Result: The decision tree was the best-performing model, achieving a test accuracy of 88.89% in predicting successful landings.

https://github.com/ngfangkiang/Data-Science-Coursework/blob/main/Applied%20Data%20Science%20Capstone/Module%204%20-%20%20SpaceX_Machine%20Learning%20Prediction.ipynb

Results

- Exploratory data analysis results
  - The study found that payload mass, year, and launch location can affect mission outcomes.
- Interactive analytics demo in screenshots

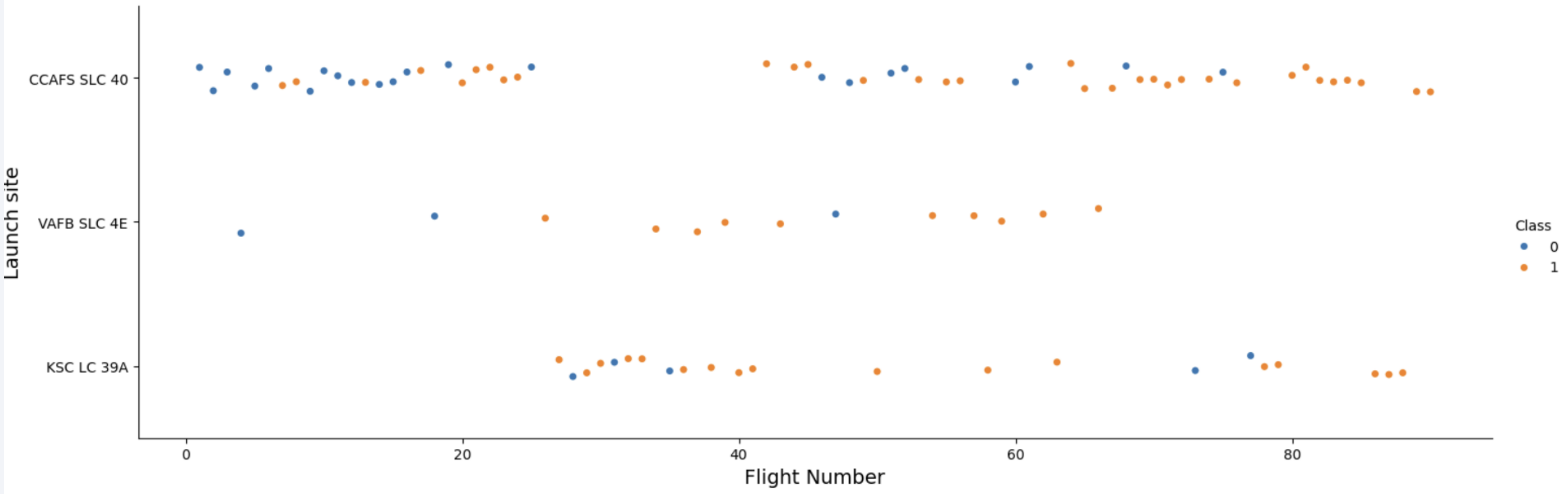




- Predictive analysis results
  - The decision tree was the best-performing model, achieving a test accuracy of 88.89% in predicting successful landings.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

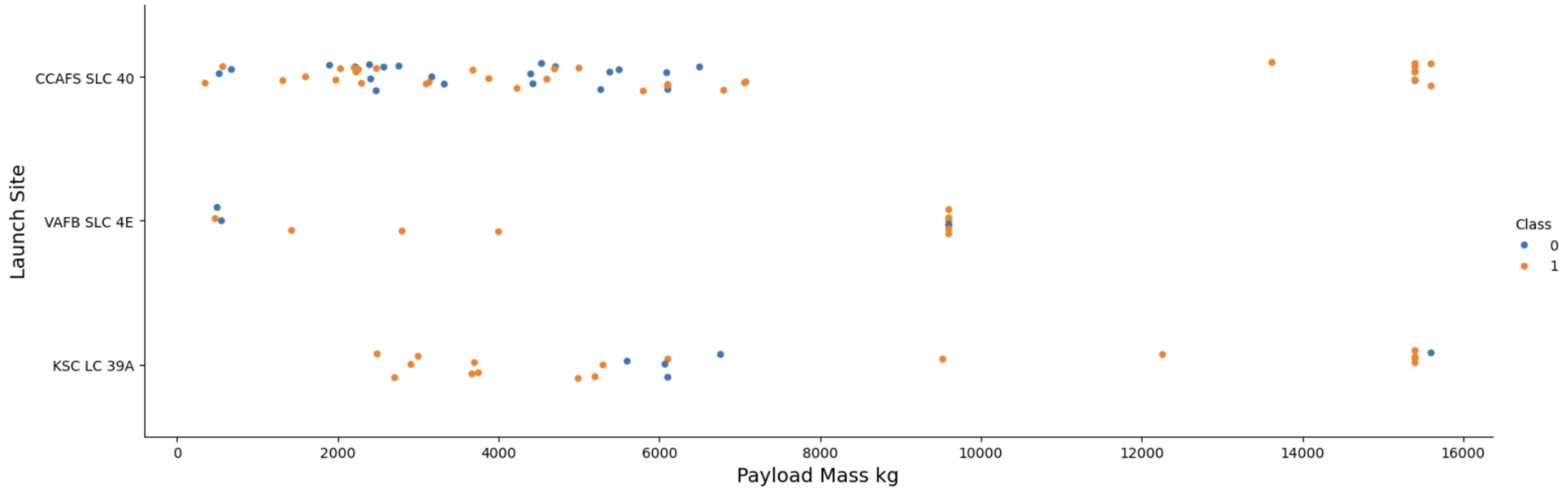


The plot reveals distinct patterns across launch sites. CCAFS SLC 40 shows a mix of successes (orange) and failures (blue) initially, with successes becoming more prevalent over higher flight numbers.

KSC LC 39A, used for later flights, exhibits a consistently high success rate.

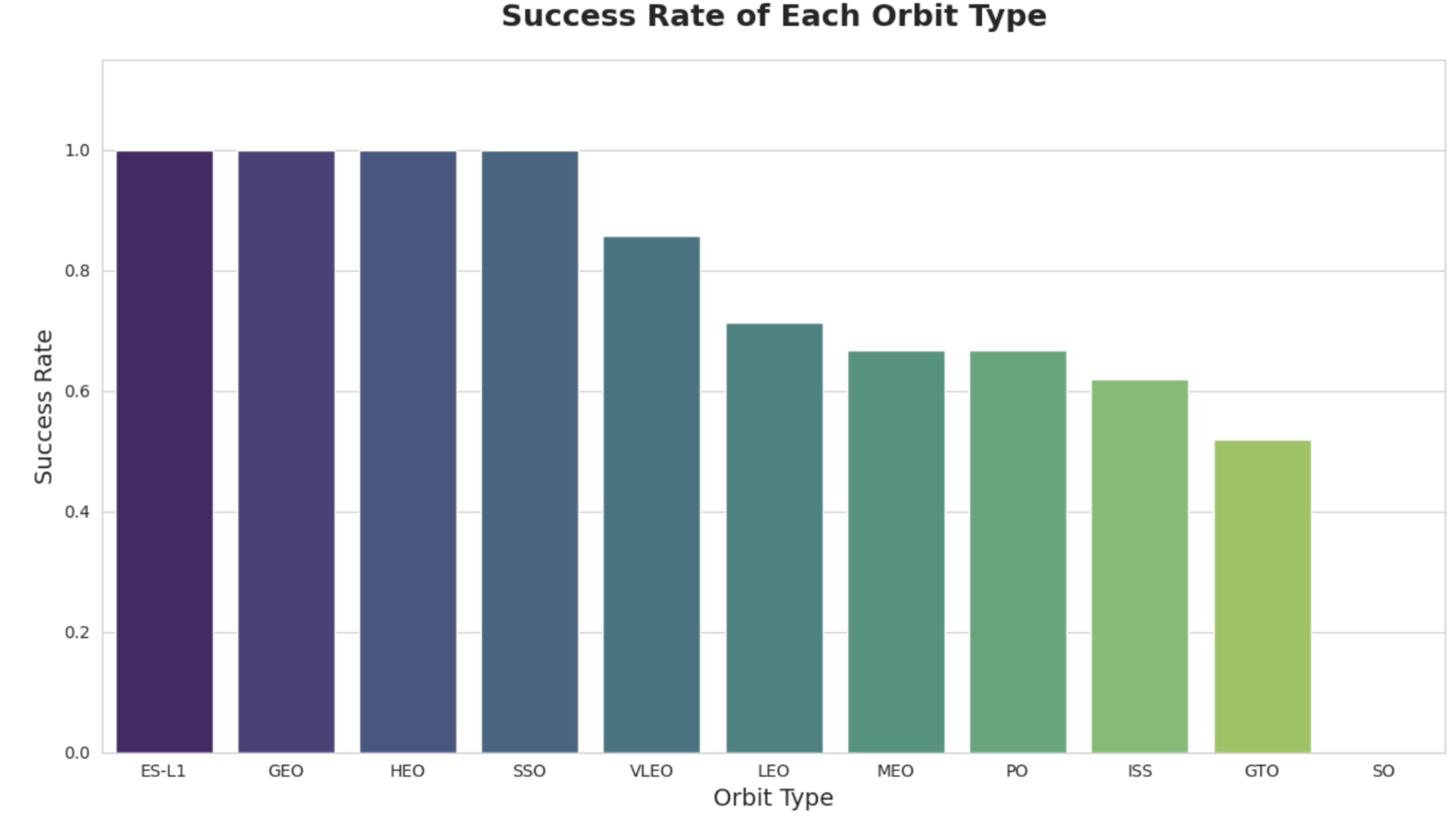
VAFB SLC 4E has very few flights, making success patterns less discernible.

Payload vs. Launch Site



Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

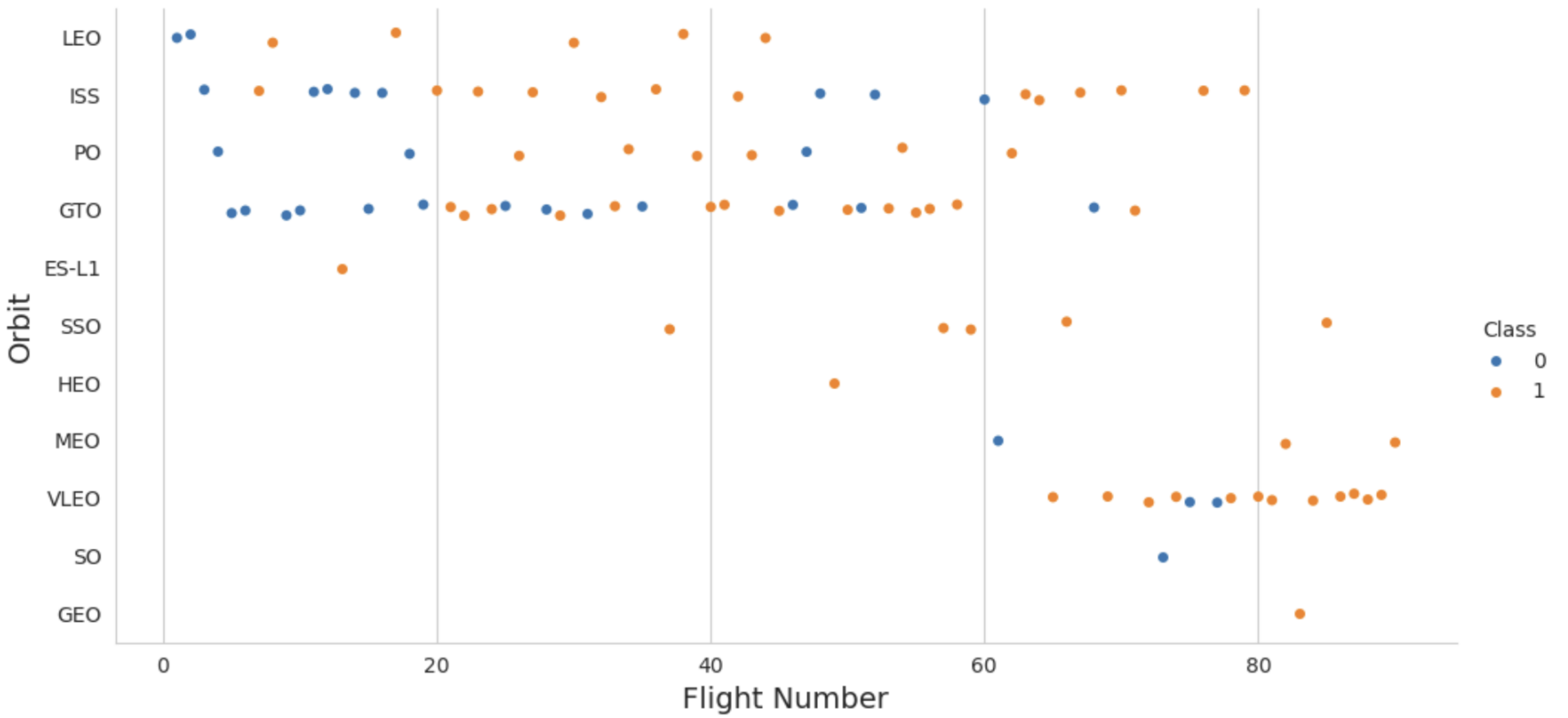
Success Rate vs. Orbit Type



Analyze the plotted bar chart to identify which orbits have the highest success rates.

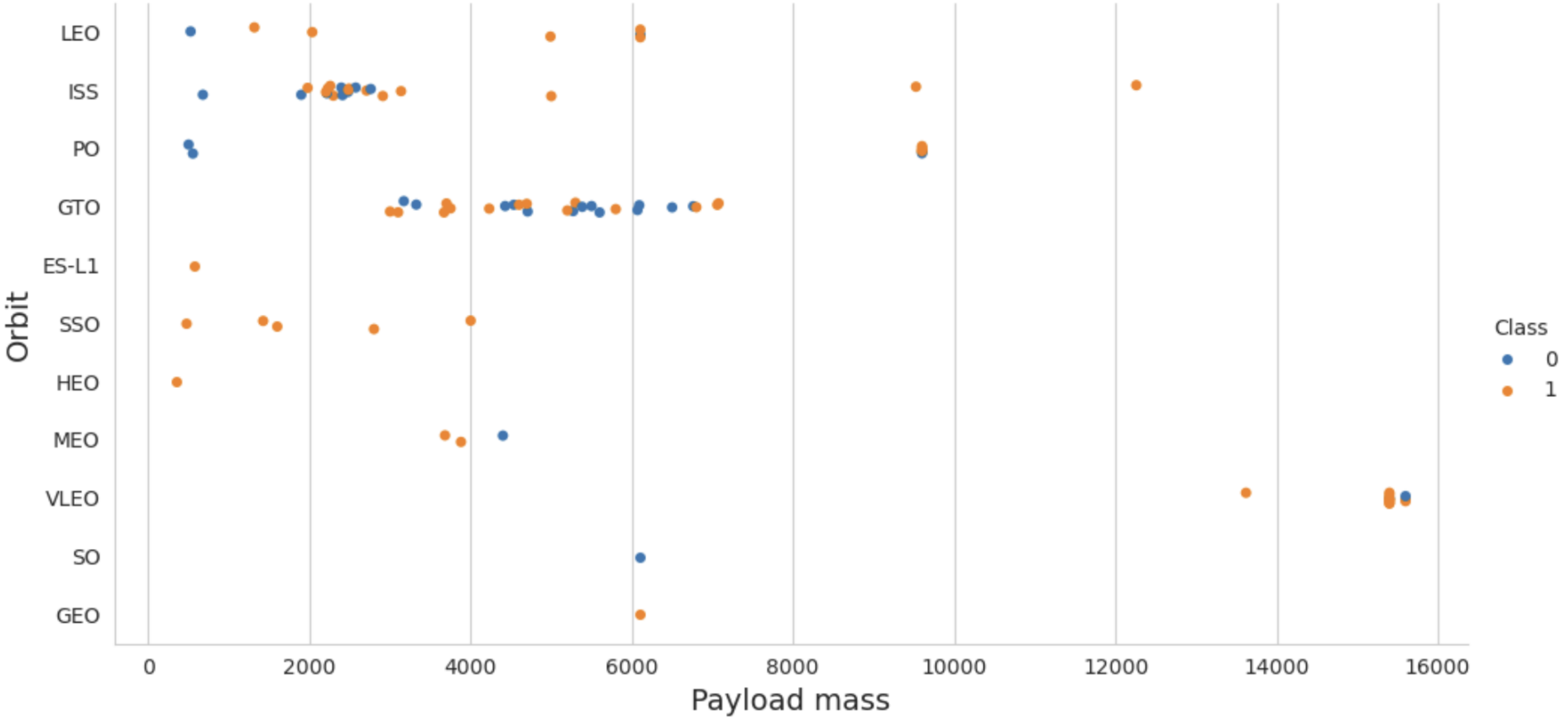
The bar chart shows ES-L1, GEO, HEO, and SSO orbits have the highest success rates, all at a perfect 1.0. Success rates generally decline for VLEO, LEO, MEO, PO, and ISS. GTO has a lower success rate around 0.5, and the SO orbit shows a 0.0 success rate.

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

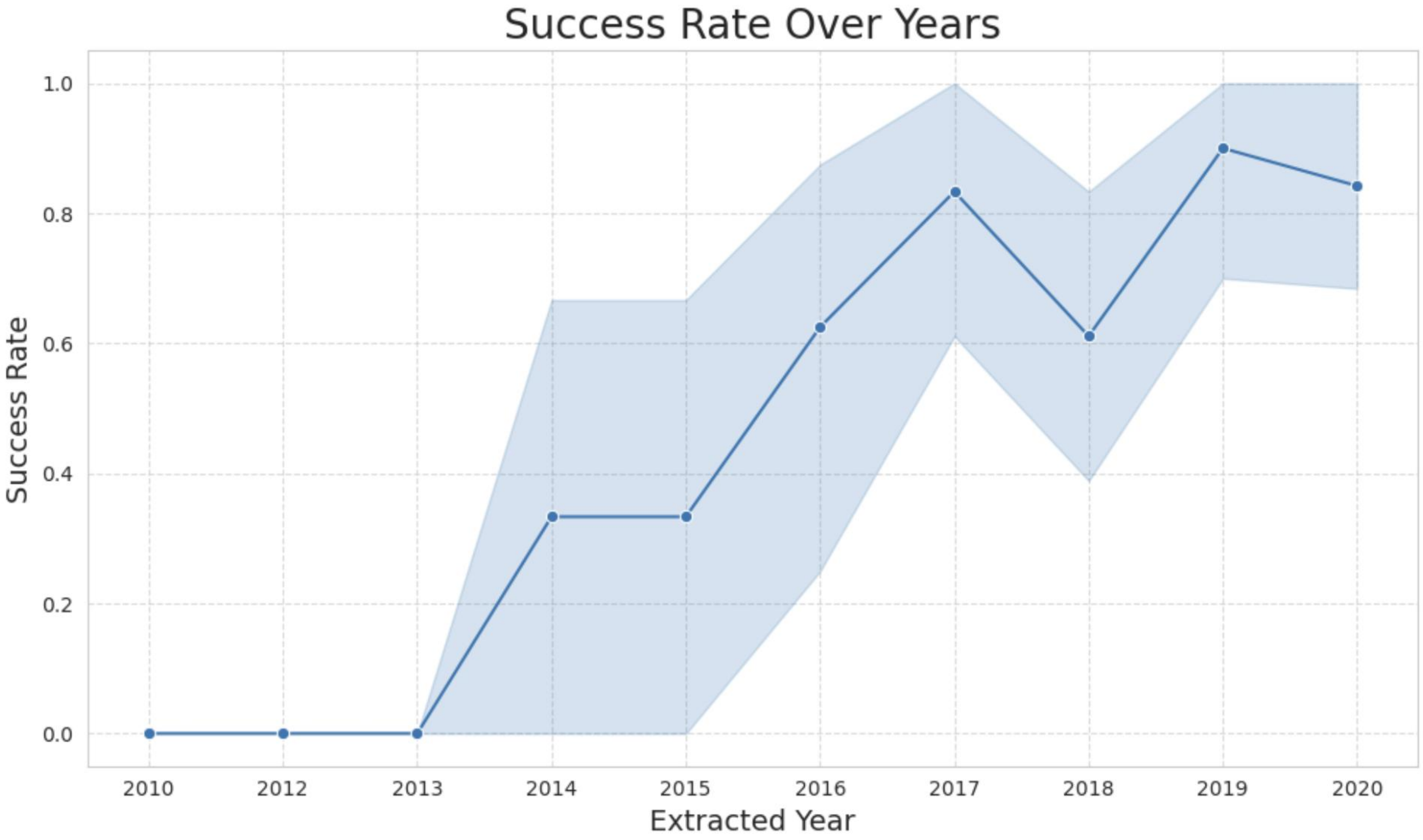
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



you can observe that the sucess rate since 2013 kept increasing till 2020

All Launch Site Names

- Find the names of the unique launch sites

Display the names of the unique launch sites in the space mission

In [10]:
```
%sql select distinct(LAUNCH_SITE) from SPACEXTBL
```

```
 * sqlite:///my_data1.db
Done.
```

Out[10]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`

Display 5 records where launch sites begin with the string 'CCA'

In [11]:
```
%sql select * from SPACEXTBL where LAUNCH_SITE like 'CCA%' limit 5
```

```
 * sqlite:///my_data1.db
Done.
```

Out[11]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

Display the total payload mass carried by boosters launched by NASA (CRS)

In [12]:
```
%sql select sum(PAYLOAD_MASS__KG_) from SPACEXTBL where CUSTOMER = 'NASA (CRS)'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[12]:

sum(PAYLOAD_MASS__KG_)
45596

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

Display average payload mass carried by booster version F9 v1.1

In [13]:
```
%sql select avg(PAYLOAD_MASS__KG_) from SPACEXTBL where BOOSTER_VERSION = 'F9 v1.1'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

avg(PAYLOAD_MASS__KG_)
2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
In [15]: %sql select min(DATE) from SPACEXTBL where Landing_Outcome = 'Success (ground pad)'

 * sqlite:///my_data1.db
Done.
```

Out[15]:

min(DATE)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
In [16]: %sql select BOOSTER_VERSION from SPACEXTBL where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS__KG_ > 4000 and I
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

List the total number of successful and failure mission outcomes

In [17]:
```
%sql select count(MISSION_OUTCOME) from SPACEXTBL where MISSION_OUTCOME = 'Success' or MISSION_OUTCOME = 'Failure (in fligh
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

count(MISSION_OUTCOME)
99

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

In [18]:

```
%sql select BOOSTER_VERSION from SPACEXTBL where PAYLOAD_MASS__KG_ = (select max(PAYLOAD_MASS__KG_) from SPACEXTBL)
```

```
 * sqlite:///my_data1.db
Done.
```

Out[18]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

In [19]:

```
%sql SELECT substr(Date, 6,2), Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome = "Failure (drone ship)"
```

```
 * sqlite:///my_data1.db
Done.
```

Out[19]:

substr(Date, 6,2)	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [20]:

```
%sql SELECT Landing_Outcome, COUNT(Landing_Outcome) FROM SPACEXTABLE WHERE Date BETWEEN "2010-06-04" AND "2017-03-20" GROUP
```

```
 * sqlite:///my_data1.db
Done.
```

Out[20]:

Landing_Outcome	COUNT(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

Locations of Launch Sites

- Are all launch sites in proximity to the Equator line?

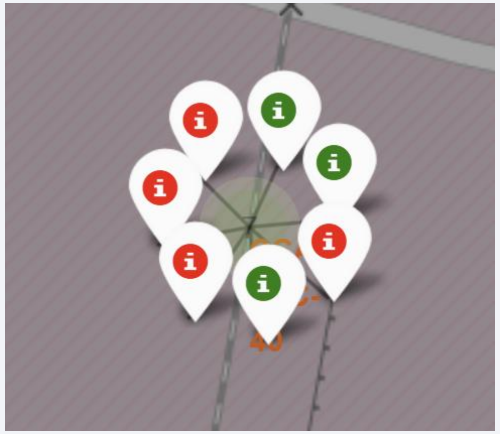
No. The two launch sites shown in the image, Vandenberg Space Force Base (VAFB) in California and Kennedy Space Center/Cape Canaveral Space Force Station (KSC/CCAFS) in Florida, are both located in the mid-latitudes of the Northern Hemisphere, significantly north of the Equator.

- Are all launch sites in very close proximity to the coast?

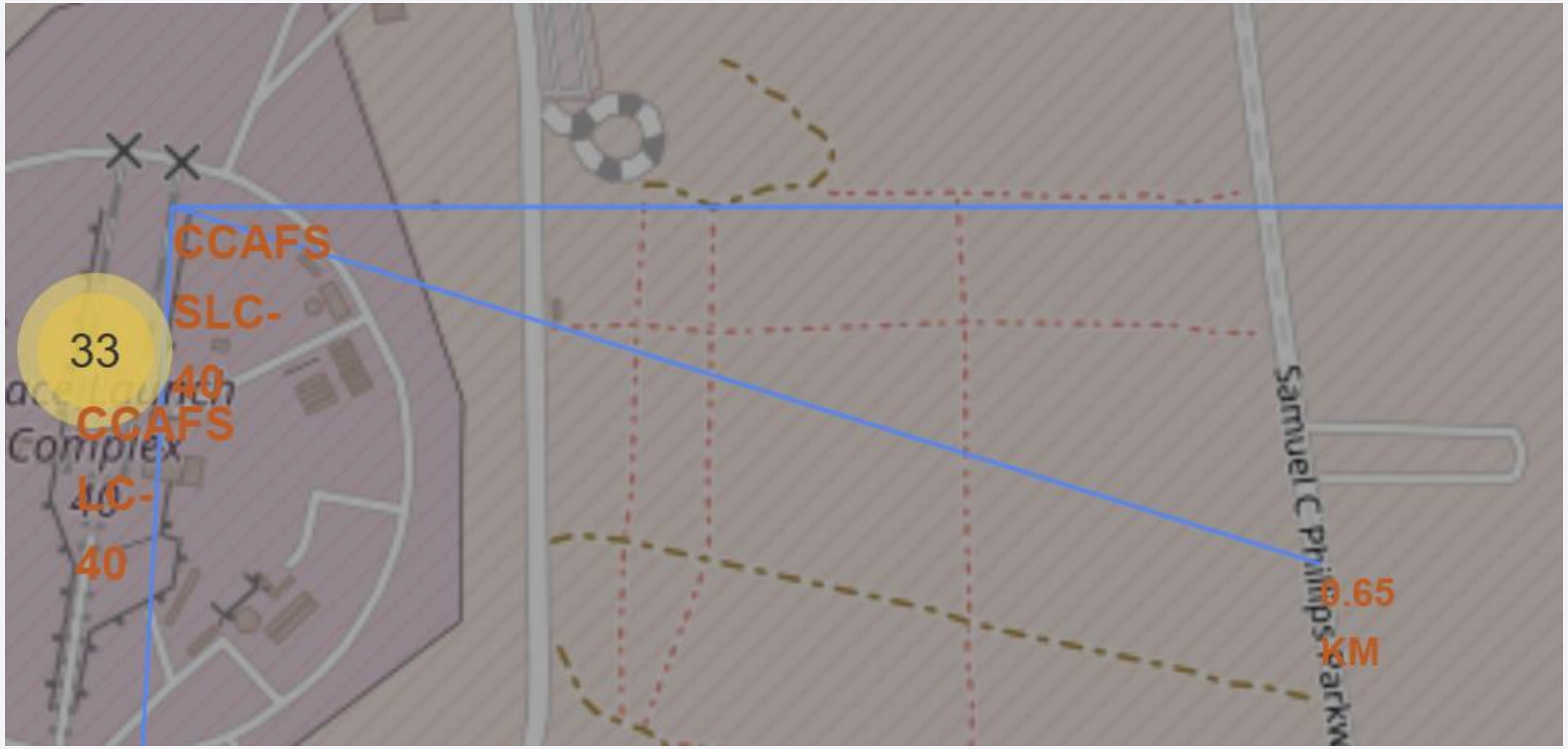
Yes. Both launch sites depicted in the image are located directly on the coast. VAFB is on the Pacific coast, and KSC/CCAFS is on the Atlantic coast. This is a common characteristic for many launch sites globally, as it allows rockets to launch over open water, minimizing risk to populated areas from falling debris or in case of a launch anomaly.

Success/failed Launches for Each Site on the Map

- Color-coded map markers (green=success, red=failure) show SpaceX launch outcomes.
- This site experienced a mix of 3 successful and 4 unsuccessful launches, indicating varied performance.



Distance from launch site to Highway



- Are launch sites in close proximity to highways?

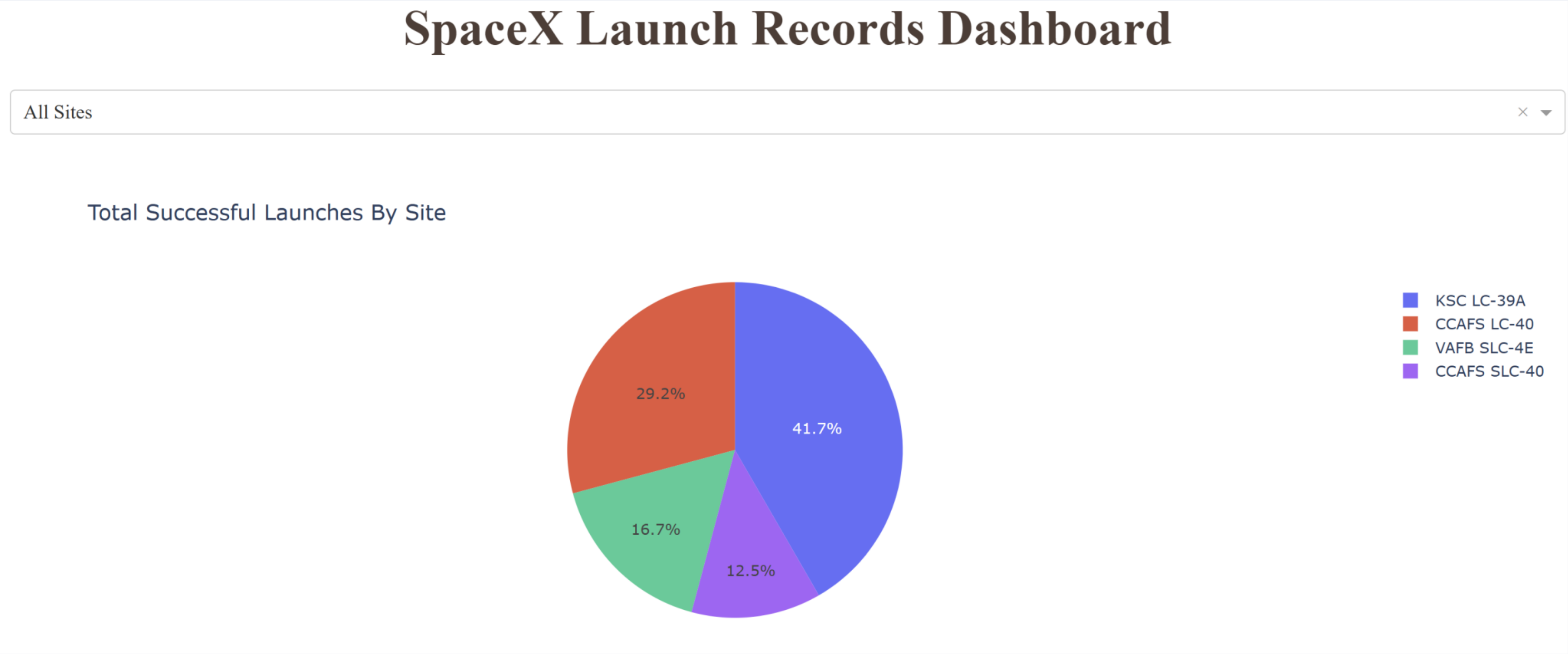
Yes, with a distance of approximately 0.65 km (650 meters), launch sites appear to be in close proximity to highways.

Section 4

Build a Dashboard with Plotly Dash

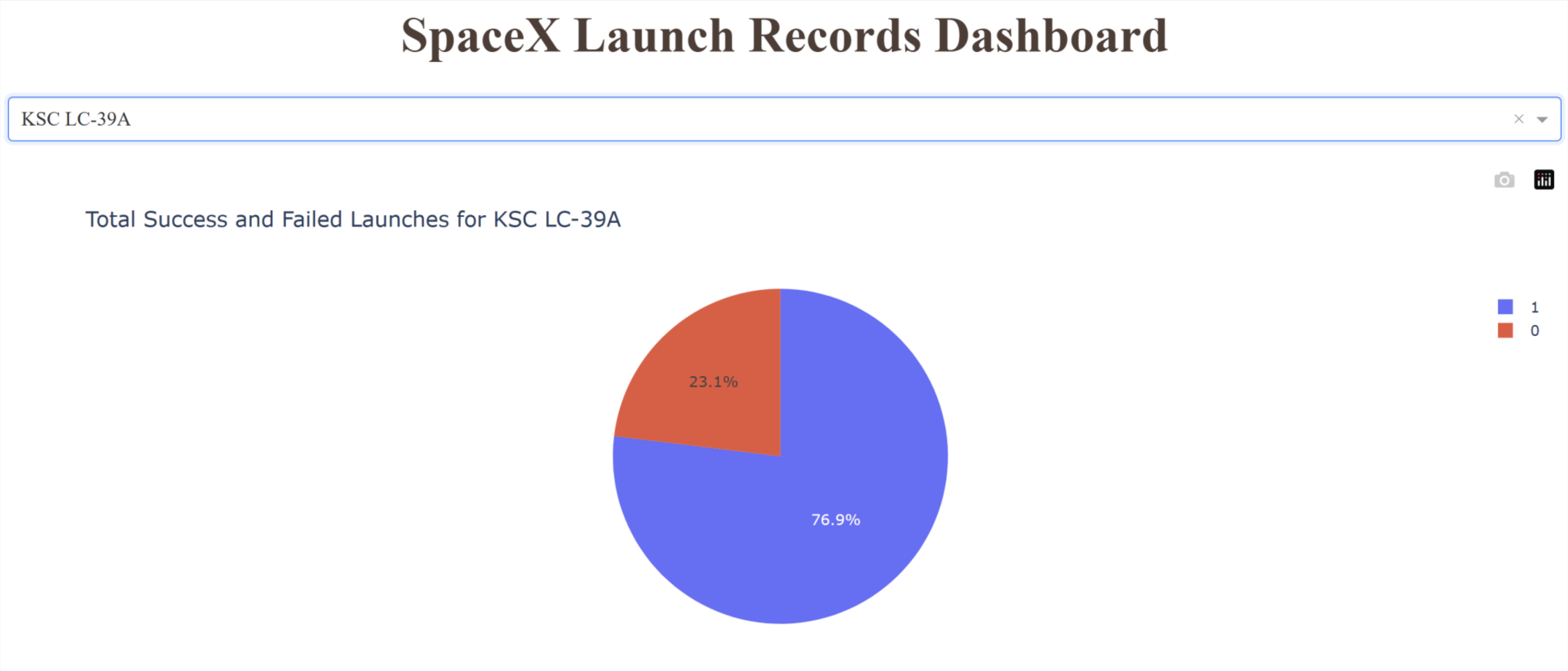
Distribution of Successful Launches Across Sites

- KSC LC-39A is the site with the most successful launches.



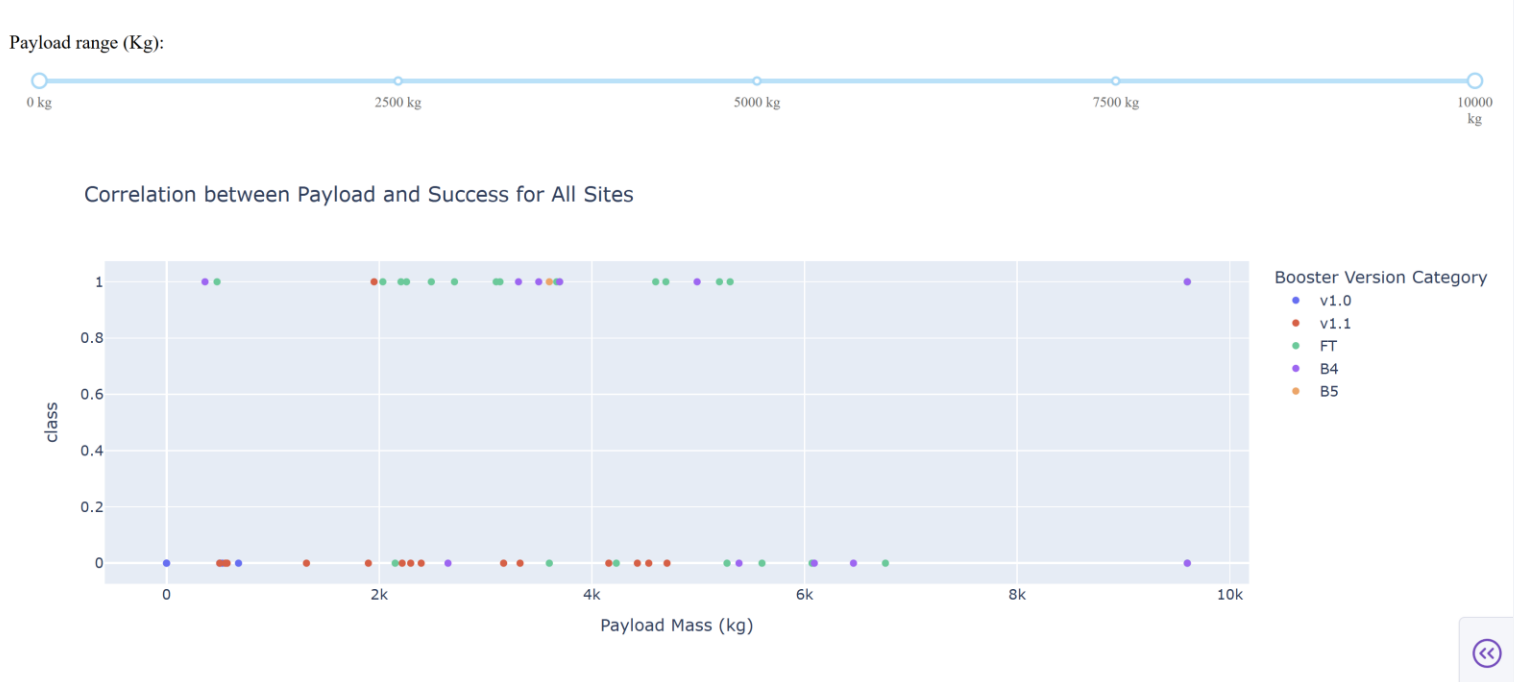
<Dashboard Screenshot 2>

- A high launch success ratio is observed at the KSC LC-39A launch site, where 76.9% of launches are successful compared to 23.1% that fail.



Payload Mass vs. Launch Outcome for All Sites

- This plot shows launch success (1) vs. failure (0) by payload mass and booster.
- Payload masses (above ~2000kg and below ~5300kg) correlate with greater success.
- Newer booster versions (FT, B4, B5) show significantly higher success rates.
- Early versions (v1.0, v1.1) had more failures, especially at lower masses.

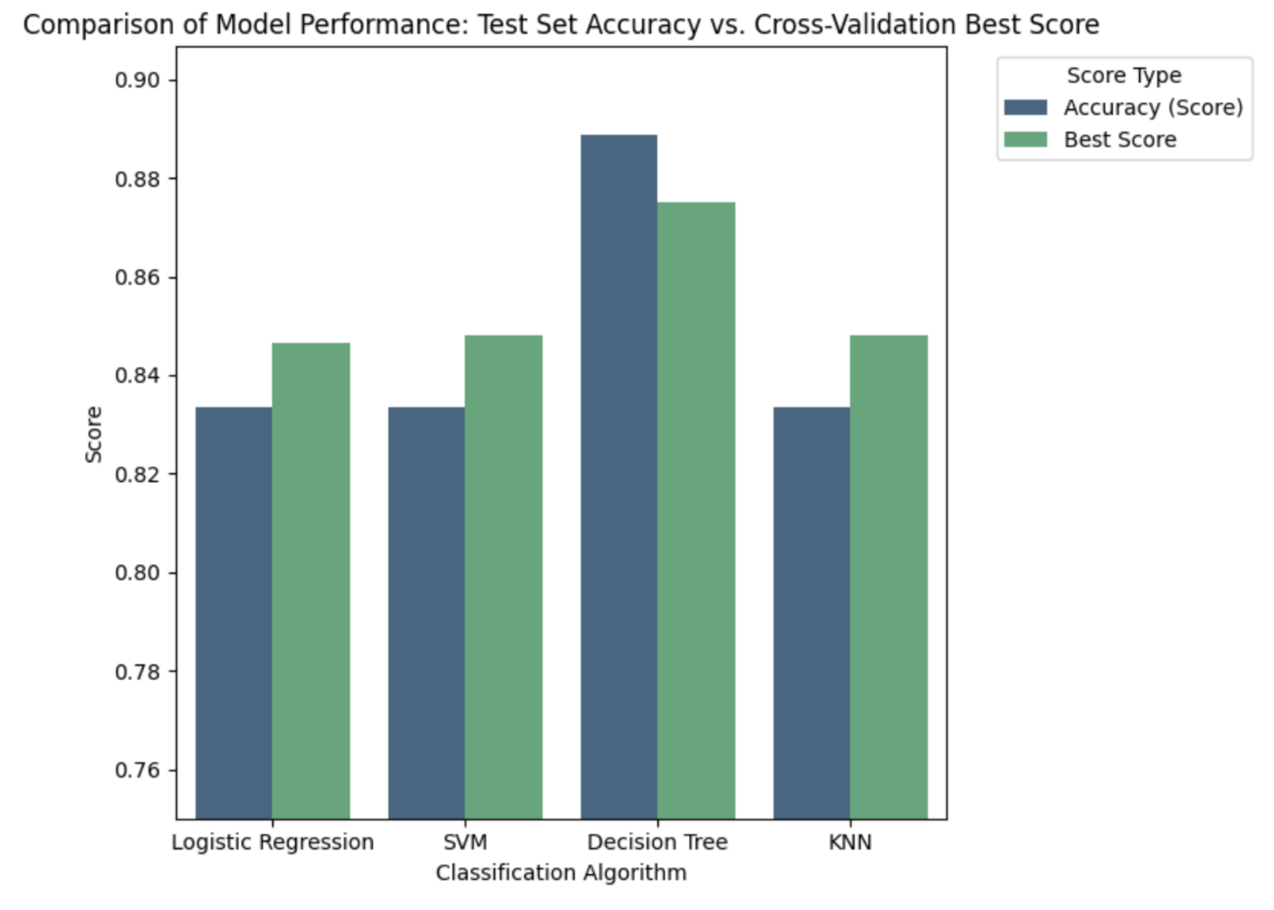


Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Based on the "Accuracy (Score)" (blue bars) in the chart, the **Decision Tree** model has the highest classification accuracy, with a score close to 0.89.



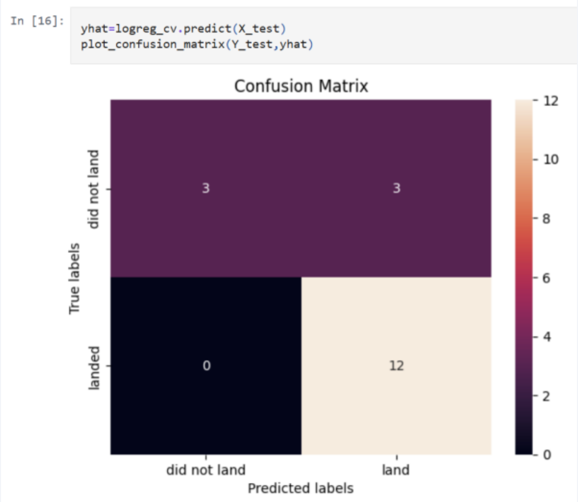
	Model	Score Type	Score Value
0	Logistic Regression	Accuracy (Score)	0.833333
1	SVM	Accuracy (Score)	0.833333
2	Decision Tree	Accuracy (Score)	0.888889
3	KNN	Accuracy (Score)	0.833333
4	Logistic Regression	Best Score	0.846429
5	SVM	Best Score	0.848214
6	Decision Tree	Best Score	0.875000
7	KNN	Best Score	0.848214

Confusion Matrix

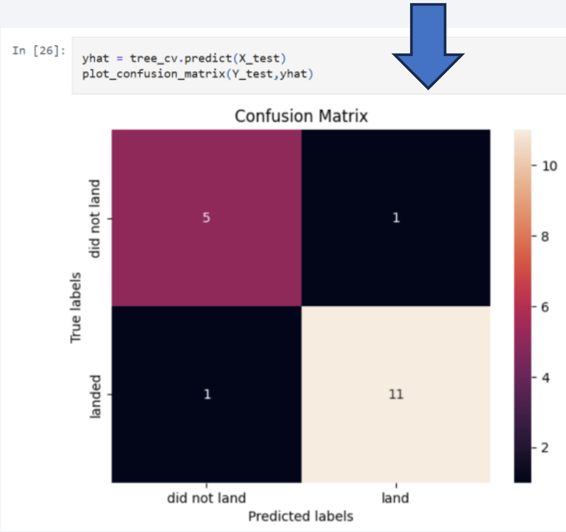
Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

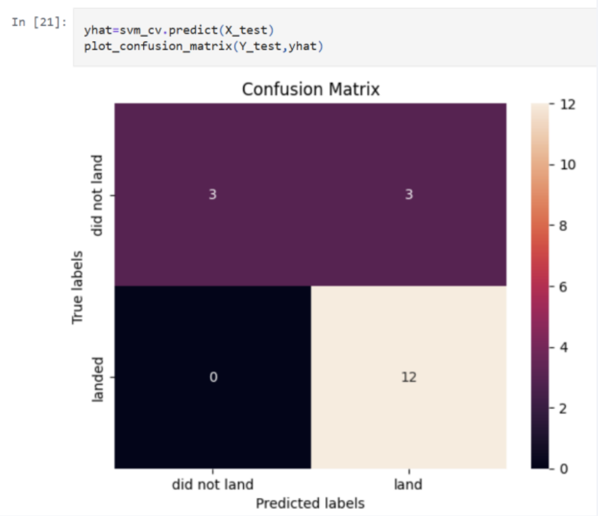
- Comparing the accuracies, the decision tree (tree_ cv) model performs best with an accuracy of 88.9%.
  - Highest Accuracy: It achieved 88.9% accuracy, outperforming all other models.
  - Fewest Errors: It made only 2 incorrect predictions, compared to 3 by the others.
  - Superior Performance: This resulted in the highest number of correctly classified instances



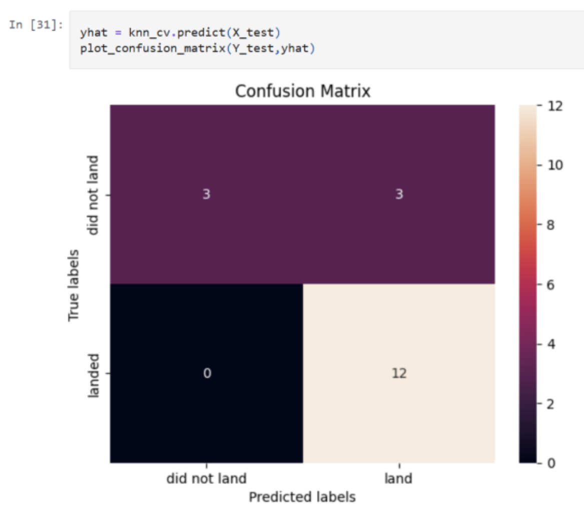
Logistic Regression



Decision Tree



SVM



KNN

Conclusions

- SpaceX's launch success rate demonstrated a consistent upward trend from 2013-2020.
- Kennedy Space Center's Launch Complex 39A (KSC LC-39A) exhibited the highest success rate among all evaluated launch sites.
- Optimal launch success was observed for payload masses between approximately 2,000 kg and 5,300 kg.
- ES-L1, Geostationary (GEO), Highly Elliptical (HEO), and Sun-Synchronous (SSO) orbits consistently yielded high success rates, with Very Low Earth Orbit (VLO) and International Space Station (ISS) orbits performing well for heavy payloads.
- Our predictive analysis identified the decision tree model as the most effective, achieving an accuracy of 88.9%.

Thank you!